AICTE Training and Learning (ATAL) Academy Sponsored ONLINE Faculty Development Programme on

"Bridging Artificial Intelligence, Quantum Computing, and Emerging Materials: Theory, Applications, and Innovation"

18th to 23rd January 2027



Organized by
Department of Mechanical Engineering
jointly
Department of Computer Science and Engineering

National Institute of Technology Silchar, Assam-788010

ABOUT THE INSTITUTE

National Institute of Technology, Silchar is one of the 31 National Institutes of Technology of India and was established in1967 as a Regional Engineering College in Assam.

In 2002, it was upgraded to the status of National Institute of Technology and was declared as Institute of National Importance under the National Institutes of Technology Act, 2007. NIT Silchar is a fully residential campus situated on the banks of river Barak and on a sprawling campus spread over 625 acres of land surrounded by scenic tea gardens on the outskirts of Silchar.

NIT Silchar has secured 50 rank in Engineering and 97 rank under Overall Category in NIRF 2025. Green Metric Global Rank in 2025 is 191 and QS Asia Ranking 2026 is 552.

SPONSORED BY

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Dept. of CSE, NIT Silchar

Organizing Committee

Faculty members,
Dept. of Mechanical and Computer Science and Engineering,
NIT Silchar, Assam, Pin:788010

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ABOUT THE DEPARTMENT

The Department of Mechanical Engineering is one of the oldest as well as finest in the zone and carries a rich legacy from thereon. From the very onset, the main objective of the department is to strengthen the academic and technical knowledge of the students through various world-class technologies, teaching-learning, and research activities. Over these glorious years of its journey, the department is keeping itself up to date with the latest technological and sustainable developments and trends in the field and with a dedicated faculty of highly qualified and experienced members in all streams of mechanical engineering. The curriculum is revised continuously with the active involvement of industry, alumni, research organizations, and other stakeholders, and UG and 03 numbers of PG programs are accredited by the National Board of Accreditation (NBA).

The department of Computer Science and Engineering offers B. Tech. in CSE, M. Tech. with three specialization CSE, AI, Data Science & Engineering and Ph. D. programmes. With sound Vision and Mission, this department provides a platform for nurturing Engineers and researchers to lead the country in particular and the entire World in general. The department is well-equipped with ultra-modern facilities for research. To name a few the department has Blade Server (IBM 7 blades), Thermal Cameras, Workstations, latest configured Desktop PCs, etc. Furthermore, the department has procured dozens of software, namely, IBM Rational Rose, QUALNET, Red-hat linux, Windows, Matlab, etc. The faculties and the students of the department work in tandem to create a unique identity in the country.

ABOUT THE EVENT

The objective of the Faculty Development Programme (FDP) on "Bridging Artificial Intelligence, Quantum Computing, and Emerging Materials: Theory, Applications, and Innovation" is to establish a direct link between the design, constraints, and architectures of physical qubit devices and the physics, synthesis, and characterization of quantum materials (such as superconductors and topological insulators). To teach participants about the logical levels of quantum computing, such as variational techniques, quantum error correction, fundamental algorithms (such as Grover's and Shor's), and immediate applications like quantum material simulation.

TOPICS TO BE COVERED

- Recent Trend on Quantum Computing
- Quantum Material and Technology
- Quantum Information Theory and AI
- Quantum Material and Application
- Quantum NLP
- Application Quantum Technology in Industry

RESOURCE PERSONS

Industry Experts:

Mr. Ujjwal Sonkar, Co-Founder & CEO, Quantumshift, India
Dr. Lucas Ondel Yang, Laboratoire Interdisciplinaire des Sciences du Numérique, France
Dr. Anjani Priyadarsini, Quantum Industry Specialist, Former Amazon (US)
Dr. Prajnesh Kumar, Quantum Technology Lead, Quantum Computing Inc. USA
Dr. Benoit Favre, Naver Labs, Grenoble, France
Mr. Mainal Ray, Director of Operations Radio Design India Pvt Ltd, India

Overseas Experts:

Dr. Lucas Ondel Yang,Laboratoire Interdisciplinaire des Sciences du Numérique, France
Dr. Anjani Priyadarsini, Quantum Industry Specialist, Former Amazon (US)
Dr. Prajnesh Kumar, Quantum Technology Lead, Quantum Computing Inc. USA
Dr. Benoit Favre, Naver Labs, Grenoble, France

National Experts:

Dr. Vivek Kumar Rai, Former Scientist F, DST TBD
Prof. Amlan Chakrabarti, Calcutta University, India
Prof. Goutam Paul, ISI, Kolkata, India
Prof. Sivaji Bandyopadhyay, Jadavpur University, India
Dr. Arijit Mondal, IIT Kharagpur, India
Prof. Kaushik Kumar, Professor, BIT Mesra
Mr. Mainal Ray, Director of Operations Radio Design India Pvt Ltd, India

Important Dates

Event Dates: **18th to 23rd January 2027**

Eligibility

Open to Faculty, Research scholars and PG students.

Registration fees: Nil

No.of seats: 100+

Online Registration link:
Will be updated after acceptance